The dressing spectrum from chords

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`syk_gravity` project

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Abstract

The previous report reduced the decoder's localization to the distribution of *dressing costs*: the eigenvalues of the operator $\mathcal{D} = P\,E^\dagger H_0^{-1} E\,P$ on the BPS space of the smaller model, where E is the pair-creation operator of the added mode. $\mathcal{D}$ is a Euclidean-time-integrated, zero-temperature two-point function of a random two-fermion matter operator. Here we compute it in the double-scaled chord theory of $\mathcal{N} = 2$ SYK. Three results are exact. (i) The BPS-to-continuum two-point function of a charged single-chord operator is the orthogonality measure of the Al-Salam–Chihara polynomials, with parameters fixed by the two charges and the operator's dimension. Its discrete part is the BPS-to-BPS result of Boruch, Lin and Yan; its Schwarzian limit is the $\mathcal{N} = 2$ JT result of Lin, Maldacena, Rozenberg and Shan. (ii) The time integral, i.e. the mean dressing cost, is a ${}_2\phi_1$. (iii) In the chord theory, uncrossed matter-chord diagrams make the full dressing spectrum a free compound Poisson law. With no fitted parameters, this gravitational answer reproduces the exact first-order heterogeneity $W^{(1)}$ of the decoder to 4–9% for $N' = 10$–16, and the decoder's own W to -8% to $+11\%$. We give a complete error budget. Crossed diagrams contribute at most 0.5% of the third moment. The free law is 3–5% high because the exact spectrum has a softer lower edge. The chord inputs add 1–3%. Predictions for the next size, $N' = 17$, are pre-registered. Finally, the BPS states are *not* equivalent at finite N, as the chord theory would have them. We show by an exact sum rule that this heterogeneity is the microstate dependence of single-fermion BPS-to-BPS weights. It has an incoherent part, which shrinks with N, and a coherent part, which does not.

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1 Introduction

1.1 Where we stood

The decoder of the uplift problem compares a BPS state of the N'-mode model of one-flavor $\mathcal{N} = 2$ SYK with the BPS space of the $(N' - 1)$-mode model [1, 2]. Its moments $T_{2m} = \langle y^m \rangle$, with $y = \cos^2 2\theta$ and θ the principal angle, obey the exact identity $T_4 = T_2 - W$, where $W = \langle y(1 - y) \rangle$ is the weight at intermediate angles [2, §9]. Localization, the excess of the decoder's T_4 over an isotropic deformation with the same T_2, is therefore a statement about W.

The previous report [2] found an exact uplift map and, from it, the first-order decoder. To first order an old BPS state h is dressed by the cloud $-\varepsilon K h$, and its principal angle obeys $\tan^2 \theta = \sigma^2$. Here σ^2 is an eigenvalue of the *dressing operator*

$$\mathcal{D} = P\, E^\dagger H_0^{-1} E\, P = \int_0^\infty d\tau\; P\, E^\dagger e^{-\tau H_0} E\, P, \tag{1.1}$$

acting on the old BPS space at charge F (Prop. 5 of [2]). H_0 is the old Hamiltonian, and $E = \eta\wedge = \sum_I \eta_I \psi_I^\dagger$ creates a pair of fermions with the couplings η of the added mode. The first-order heterogeneity $W^{(1)}$, computed from these dressing costs, reproduced the decoder's W to 1–7% for $N' = 11$–15. It also passed a pre-registered test at $N' = 16$. The report closed by noting that $\mathcal{D}$ is a time-integrated zero-temperature two-point function of a random two-fermion matter operator. Its spectrum is therefore the kind of quantity the chord formalism of BLY [4] and BINT [6] computes.

1.2 What this report does

We carry out that computation and test it.

1. **The two-point function in closed form** (§3, [derived + verified]). Theorem 1: the BPS-to-continuum spectral measure of a charged chiral single-chord operator is the Al-Salam–Chihara

measure. It contains BLY's BPS-to-BPS formula as its point mass (Corollary 1) and LMRS's super-Schwarzian formula as its triple-scaling limit (Corollary 2).

2. **The time integral** (§4, [derived + verified]). Proposition 1: the mean dressing cost is a ${}_2\phi_1$, and it grows like $\lambda^{2\Delta-2}$ as $\lambda \to 0$.

3. **The whole spectrum** (§5, [derived under the chord rules]). Wick contractions of the couplings are matter-chord diagrams. Uncrossed diagrams sum to a free compound Poisson law (Proposition 2). Crossed diagrams are measured to be at most 0.5% of the third moment [numerical].

4. **Gravity versus exact** (§6, [numerical]). The chord prediction of $W^{(1)}$ is 4–9% high at $N' = 10$–16. We split the error into its sources and identify a soft lower edge of the exact spectrum.

5. **A pre-registered test at** $N' = 17$ (§7, [pre-registered]).

6. **BPS heterogeneity** (§8). An exact sum rule (Proposition 3) ties the part of the physics the chord theory misses to single-fermion BPS-to-BPS weights.

2 Setup

2.1 The dressing problem

The old model has $N = N' - 1$ complex fermions and supercharge $Q_0 = \omega_0 \wedge$ with a random 3-form ω_0 ($p = 3$, $E|C_{ijk}|^2 = 2$). Its BPS states at charge $F = \lfloor N'/2 \rfloor$ form the space $\mathcal{B}^F$ of dimension b_F, with projector P. There are two dressing problems [2]:

- the h **family**: $\mathcal{D} = P\,E^\dagger G\,E\,P$ on $\mathcal{B}^F$, where $G = H_0^{-1}$ acts on the e_{F+2} exact forms of degree $F+2$;
- the g **family**: $\mathcal{D}' = P'\,E\,G'E^\dagger P'$ on $\mathcal{B}^{F-1}$, where $G' = H_0^{-1}$ acts on the e'_{F-3} coexact forms of degree $F-3$.

The pair operator $E = \sum_{i<j} \eta_{ij} c_i^\dagger c_j^\dagger$ has independent couplings with $E|\eta_{ij}|^2 = 2$. The first-order decoder eigenvalues are $y = \big((1-s)/(1+s)\big)^2$ for $s \in \operatorname{spec}\mathcal{D} \cup \operatorname{spec}\mathcal{D}'$, and $W^{(1)} = \langle y(1-y)\rangle$ over all $b_F + b_{F-1}$ states. We call $s = \sigma^2$ the dressing cost. Two exact facts fix the scale:

$$\textstyle\sum_{i<j} \psi_{ij}\psi_{ij}^\dagger = \binom{N-F}{2} \text{ on } \Lambda^F \quad \Rightarrow \quad E_\eta\langle h|E^\dagger E|h\rangle = c_0 \equiv 2\binom{N-F}{2}, \tag{2.1}$$

and likewise $c_0' = 2\binom{F-1}{2}$ for the g family. The normalized two-point function is $G(\tau) = E_\eta\langle h|E^\dagger e^{-\tau H_0} E|h\rangle/c_0$, and the mean dressing cost is $t_1 = c_0 \int_0^\infty G(\tau)\,d\tau$.

2.2 Double-scaled chords at fixed charge

We use the one-sided chord Hilbert space of BBNR [3, §§4.4–5.1]. A chord state is an alternating string of chord ends, read from top to bottom:

$$|\emptyset\rangle, \qquad |n, X\rangle = XOXO\cdots XO, \qquad |n, O\rangle = OXOX\cdots OX \qquad \text{(length } 2n\text{)},$$
$$|n{+}\tfrac{1}{2}, X\rangle = XOX\cdots X, \qquad |n{+}\tfrac{1}{2}, O\rangle = OXO\cdots O \qquad \text{(length } 2n+1\text{)}.$$

At fixed charge s, Q opens an O chord at the bottom with weight q^s and closes X chords. $Q^\dagger$ opens X with q^{-s} and closes O. Non-alternating strings are null. The inner product is BBNR (4.37), and

$T = QQ^\dagger + Q^\dagger Q$. Here $q = e^{-\lambda}$ and $\lambda = 2p^2/N$. The chord charge of a fermion-number sector of degree d is $s = (d - N/2)/p$. Applied to the states we need, these rules give

$$\begin{aligned} Q|n, X\rangle &= -q^{n-1}|n-\tfrac{1}{2}, O\rangle, & Q|n, O\rangle &= q^s|n+\tfrac{1}{2}, O\rangle + q^{-1}|n-\tfrac{1}{2}, O\rangle, \\ Q|\emptyset\rangle &= q^s|\tfrac{1}{2}, O\rangle, & Q|n+\tfrac{1}{2}, O\rangle &= 0, \\ Q|\tfrac{1}{2}, X\rangle &= q^s|1, X\rangle + |\emptyset\rangle, & Q|n+\tfrac{1}{2}, X\rangle &= q^s|n+1, X\rangle + |n, X\rangle + q^n|n, O\rangle \quad (n \geq 1), \end{aligned} \tag{2.2}$$

with $Q^\dagger$ obtained by exchanging $O \leftrightarrow X$ and $s \to -s$. The norms are

$$\begin{aligned} \langle n, O|n, O\rangle &= \langle n, X|n, X\rangle = q^{-n}(q^2;q^2)_{n-1}, & \langle n, O|n, X\rangle &= -(q^2;q^2)_{n-1}, \\ \langle n+\tfrac{1}{2}, X|n+\tfrac{1}{2}, X\rangle &= q^{s-n}(q^2;q^2)_n, & \langle n+\tfrac{1}{2}, O|n+\tfrac{1}{2}, O\rangle &= q^{-s-n}(q^2;q^2)_n. \end{aligned} \tag{2.3}$$

[literature] These follow from BBNR (4.16)–(4.17) and (4.37). We checked them in a transfer-matrix implementation (`src/chord_transfer.py`), which reproduces BBNR (4.40)–(4.42) and (5.5), BLY (3.17) and BLY (4.7).

The bosonic sector splits into the T-invariant families $\mathcal{B} = Q\,\mathcal{F}$ (exact) and $\bar{\mathcal{B}} = Q^\dagger\mathcal{F}$ (coexact). For $|s| < \frac{1}{2}$ there is in addition a unique zero mode Ψ_s, the supersymmetric Hartle–Hawking state of BLY. On $\mathcal{B}$, with $b_n = Q|n+\frac{1}{2}, X\rangle$, BBNR (5.5) reads $Tb_n = q^{-1}(1-q^{2n})b_{n-1} + (q^{-s}+q^{s-1})b_n + b_{n+1}$. Its spectrum is

$$\Lambda_s(\phi) = 2q^{-1/2}\big[\cosh\lambda(s-\tfrac{1}{2}) - \cos\phi\big], \qquad \phi \in (0, \pi), \tag{2.4}$$

gapped unless $s = \frac{1}{2}$.

Remark (A sign in BBNR). The eigenvector of (2.4) is $v(\phi) = \sum_n \alpha_n(\phi) b_n$ with $\alpha_n = q^{n/2}H_n(-\cos\phi\,|\,q^2)/(q^2;q^2)_n$, where H_n are the continuous q-Hermite polynomials. BBNR (5.10) writes $H_n(+\cos\phi|q^2)$. That is the eigenvector for $+\cos\phi$ in (2.4), which can be checked directly from (5.5). The slip is harmless for everything in BBNR, because their measure is symmetric under $\phi \to \pi - \phi$. It matters for kernels with one BPS end, which are not symmetric. Combining (5.10) with (5.3) naively reflects the BPS-to-continuum weight from the bottom to the top of the band.

The normalization is $\langle\emptyset|v(\phi)\rangle = 1$ and $\langle v(\phi)|v(\phi')\rangle = q^s\Lambda_s(\phi)\,2\pi\delta(\phi-\phi')/(q^2, e^{\pm 2i\phi}; q^2)_\infty$ (BBNR 5.11). The $\mathcal{B}$-family density at charge s is therefore

$$\rho_B(\phi) = \frac{(q^2, e^{\pm 2i\phi}; q^2)_\infty\, q^{-s}}{2\pi\,\Lambda_s(\phi)}. \tag{2.5}$$

2.3 Charged matter

A chiral single-chord operator built from p' fermions (BBNR §7.1) changes the charge of the region it encloses by $\delta = p'/p$. It also inserts $\tilde{q}^{\hat{N}}$ at the junction, where $\tilde{q} = q^\delta$ and $\hat{N}|n, \cdot\rangle = n|n, \cdot\rangle$. In BLY's language the operator has dimension $\Delta = \delta/2$. For the pair operator, $p' = 2$, so $\delta = \frac{2}{3}$ and $\Delta = \frac{1}{3}$. The h family starts at $s_{\text{out}} = (F - N/2)/p$ and ends at $s_{\text{in}} = s_{\text{out}} + \delta$. The g family is its mirror image under $s \to -s$. For even N' both families have $(s_{\text{out}}, s_{\text{in}}) = (\frac{1}{6}, \frac{5}{6})$. For odd N', h has $(0, \frac{2}{3})$ and g has $(\frac{1}{3}, 1)$. In every case $s_{\text{in}} > \frac{1}{2}$: the target sector has no BPS states.

We compare with finite N at $\lambda = 2p^2/N$, where $N = N' - 1$ is the size of the old model. The energy scale κ ($H_0 \leftrightarrow \kappa T$) is fixed by the exact first moment of H_0 in the target sector, $\kappa = 2[\binom{d}{3} + \binom{N-d}{3}]/(q^{s_{\text{in}}} + q^{-s_{\text{in}}})$, with $d = F + 2$ for h and $d = F - 3$ for g. The overall normalization is the exact c_0 of (2.1). Nothing is fitted.

3 The BPS-to-continuum two-point function

Theorem 1 (Closed form). *[derived + verified] Let $|s_{\rm out}| < \frac{1}{2}$ and $s_{\rm in} = s_{\rm out} + \delta > \frac{1}{2}$, and let Ψ be the zero mode at charge $s_{\rm out}$. The normalized two-point function $G(\tau) = \langle\emptyset|e^{-\tau\kappa T_{\rm in}}\tilde{q}^{\hat{N}}|\Psi\rangle/\langle\emptyset|\Psi\rangle$ is*

$$G(\tau) = \int_0^\pi d\phi\, f(\phi)\, e^{-\tau\kappa\Lambda_{s_{\rm in}}(\phi)}, \qquad f(\phi) = \frac{(ab;q^2)_\infty\,(q^2,e^{\pm 2i\phi};q^2)_\infty}{2\pi\,(ae^{\pm i\phi},\,be^{\pm i\phi};q^2)_\infty}, \qquad a = q^{s_{\rm in}-\frac{1}{2}}, \quad b = q^{\frac{1}{2}+\delta-s_{\rm out}}. \tag{3.1}$$

This is the normalized orthogonality measure of the Al-Salam–Chihara polynomials in $x = \cos\phi$, with base q^2 and parameters (a, b) [7]. Equivalently $f = \rho_B\, A$, with the squared matrix element

$$A(\phi) = \frac{\langle v(\phi)|\tilde{q}^{\hat{N}}|\Psi\rangle}{\langle\emptyset|\Psi\rangle} = \frac{(q^{2\delta};q^2)_\infty}{(q^{s_{\rm in}+\frac{3}{2}}e^{\pm i\phi},\; q^{\frac{1}{2}+\delta-s_{\rm out}}e^{\pm i\phi};q^2)_\infty}. \tag{3.2}$$

The proof has four steps.

Step 1: the BPS state. Write $\Psi = |\emptyset\rangle + \sum_{m\geq 1}\big(x_m|m,X\rangle + o_m|m,O\rangle\big)$ and set $x_0 = o_0 = 1$. By (2.2), $Q\Psi = 0$ and $Q^\dagger\Psi = 0$ read, coefficient by coefficient,

$$q^s o_m + q^{-1}o_{m+1} - q^m x_{m+1} = 0, \qquad q^{-s}x_m + q^{-1}x_{m+1} - q^m o_{m+1} = 0 \qquad (m \geq 0). \tag{3.3}$$

The solution is $x_m = q^{3m/2}H_m(-\cosh\lambda(s+\frac{1}{2})\,|\,q^2)/(q^2;q^2)_m$ (and o_m is the same with $s \to -s$). These are BLY's amplitudes (3.6)–(3.7) in BBNR's normalization. We checked them against the zero mode of the transfer matrix to 10^{-14}.

Step 2: the matter overlap. Opening an X chord carries q^{-s}, so $Q^\dagger_{\rm in} = q^{-\delta}\,Q^\dagger_{\rm out,open} + Q^\dagger_{\rm close}$. The closing part does not depend on the charge. Let $Q^\dagger_{\rm in}$ act on $\tilde{q}^{\hat{N}}\Psi$. The coefficient of $|m{+}\frac{1}{2},X\rangle$ receives three contributions: the opening term from $|m,X\rangle$, and the closing terms from $|m{+}1,X\rangle$ and $|m{+}1,O\rangle$. Using the second relation in (3.3), which is $Q^\dagger_{\rm out}\Psi = 0$, it equals $\tilde{q}^m x_m\,(q^{-s_{\rm in}} - \tilde{q}\,q^{-s_{\rm out}})$. Multiplying by the norm (2.3) gives

$$\langle b_m|\tilde{q}^{\hat{N}}|\Psi\rangle = \langle m{+}\tfrac{1}{2},X|Q^\dagger_{\rm in}\tilde{q}^{\hat{N}}\Psi\rangle = (1-\tilde{q}\,q^\delta)\,(\tilde{q}/q)^m\,(q^2;q^2)_m\,x_m. \tag{3.4}$$

A BPS state is annihilated by $Q^\dagger$. That is what turns a two-dimensional sum over chord states into this single sequence.

Step 3: q-Mehler. Combine (3.4) with $\alpha_m(\phi)$ and set $\tilde{q} = q^\delta$:

$$A(\phi) = (1-q^{2\delta})\sum_{m\geq 0}\frac{H_m(-\cos\phi|q^2)\,H_m(-\cosh\lambda(s_{\rm out}+\frac{1}{2})|q^2)}{(q^2;q^2)_m}\,(q^{1+\delta})^m.$$

The q-Mehler (Rogers) formula [8, 7],

$$\sum_{m\geq 0}\frac{H_m(\cos\theta|Q)\,H_m(\cos\psi|Q)}{(Q;Q)_m}\,t^m = \frac{(t^2;Q)_\infty}{(te^{i(\pm\theta\pm\psi)};Q)_\infty},$$

applies with $Q = q^2$, $t = q^{1+\delta}$, $e^{i\theta} = -e^{-i\phi}$ and $e^{i\psi} = -q^{s_{\rm out}+\frac{1}{2}}$. The series converges because $tq^{-s_{\rm out}-\frac{1}{2}} = b < 1$. The result is (3.2).

Step 4: a cancellation. Note that $q^{-s_{\rm in}}/\Lambda_{s_{\rm in}}(\phi) = 1/|1-ae^{i\phi}|^2$, and that the first denominator parameter of (3.2) is $q^{s_{\rm in}+3/2} = aq^2$. Hence $(1-ae^{\pm i\phi})(aq^2e^{\pm i\phi};q^2)_\infty = (ae^{\pm i\phi};q^2)_\infty$, and $f = \rho_B A$ collapses to (3.1). Since $ab = q^{2\delta}$ and $(q^{2\delta};q^2)_\infty = (1-q^{2\delta})(q^{2+2\delta};q^2)_\infty$, the prefactor is exactly the normalization of the Al-Salam–Chihara measure, $\int f\,d\phi = 1$ [7]. □

Checks. The closed form was compared with the transfer-matrix engine (exact chord rules, $\ell_{\rm max} = 200$–400). The moments $\int f\Lambda^k$ for $k = -2,\dots,3$ agree to 10^{-15} at several values of $(\lambda, s_{\rm out}, \delta)$ (`tests/test_chord_dressing.py`). The negative moments include the time integral of the next section.

Corollary 1 (BPS-to-BPS weight = BLY)**.** *[derived + verified] Continue (3.1) to $s_{\rm in} < \frac{1}{2}$, where the target sector has BPS states. Then $a > 1$, and the Al-Salam–Chihara measure acquires a point mass at $x = (a+a^{-1})/2$, i.e. at zero energy. Its weight is*

$$\frac{(a^{-2};q^2)_\infty}{(b/a;q^2)_\infty} = \frac{(q^{1-2s_{\rm in}};q^2)_\infty}{(q^{1-2s_{\rm out}};q^2)_\infty}.$$

This equals the normalized BPS-to-BPS weight of BLY (4.14), multiplied by $\sqrt{D(s_{\rm in})/D(s_{\rm out})}$ to convert between normalizations.

We checked the point mass three ways, at four parameter points, to 10^{-12}: as one minus the continuous part, as the engine's zero-mode weight, and against BLY (4.14). The discrete and continuous parts of the zero-temperature two-point function of a charged chiral operator are thus one object. BLY computed the first; Theorem 1 supplies the second.

Corollary 2 (Schwarzian limit = LMRS)**.** *[derived] Set $\phi = \lambda k$ and let $\lambda \to 0$, using $(Q^x;Q)_\infty \to (Q;Q)_\infty(1-Q)^{1-x}/\Gamma(x)$. Then*

$$f\,d\phi \;\propto\; k\sinh(\pi k)\left|\Gamma\big(\Delta - \tfrac{1}{4} + \tfrac{s_{\rm out}}{2} + \tfrac{ik}{2}\big)\,\Gamma\big(\Delta + \tfrac{1}{4} - \tfrac{s_{\rm out}}{2} + \tfrac{ik}{2}\big)\right|^2 dk. \tag{3.5}$$

This is the zero-energy-to-continuum term (43) of LMRS [5], with $j' = s_{\rm out}$, $j = s_{\rm in}$ and $s = k/2$.

To see the match, use $|\Gamma(z+1)|^2 = |z|^2|\Gamma(z)|^2$ with $z = \Delta - \frac{1}{4} + \frac{s_{\rm out}}{2} + \frac{ik}{2}$, and note that $|z|^2 \propto \Lambda$ in this limit. LMRS's measure $s\sinh 2\pi s/E_{s,j}$ times their $\Gamma(\Delta + \frac{1}{2} \pm \frac{2j'+1}{4} \pm is)$ is then (3.5). Theorem 1 is thus the q-deformation of the $\mathcal{N}=2$ JT gravity answer, just as BLY found for the BPS-to-BPS piece.

Literature. BLY [4] computed the BPS-to-BPS weights (their §4). BBNR (7.17) [3] give the moments between ground states and the continuum, integrated over the charge. We have not found in BBNR, BLY or LMRS the fixed-charge measure (3.1), its Al-Salam–Chihara form, or the time integral of the next section.

4 The time-integrated two-point function

Proposition 1 (Mean dressing cost)**.** *[derived + verified] Under the hypotheses of Theorem 1,*

$$\int_0^\infty G(\tau)\,d\tau = \frac{1}{\kappa}\int_0^\pi \frac{f(\phi)}{\Lambda_{s_{\rm in}}(\phi)}d\phi = \frac{1}{\kappa}\,\frac{q^{s_{\rm in}}}{1-q^{2\delta}}\;{}_2\phi_1\Big(q^2,q^2;\,q^{2+2\delta};\,q^2,\,q^{2s_{\rm in}-1}\Big), \tag{4.1}$$

and the mean first-order dressing cost is $t_1 = c_0\int_0^\infty G\,d\tau$.

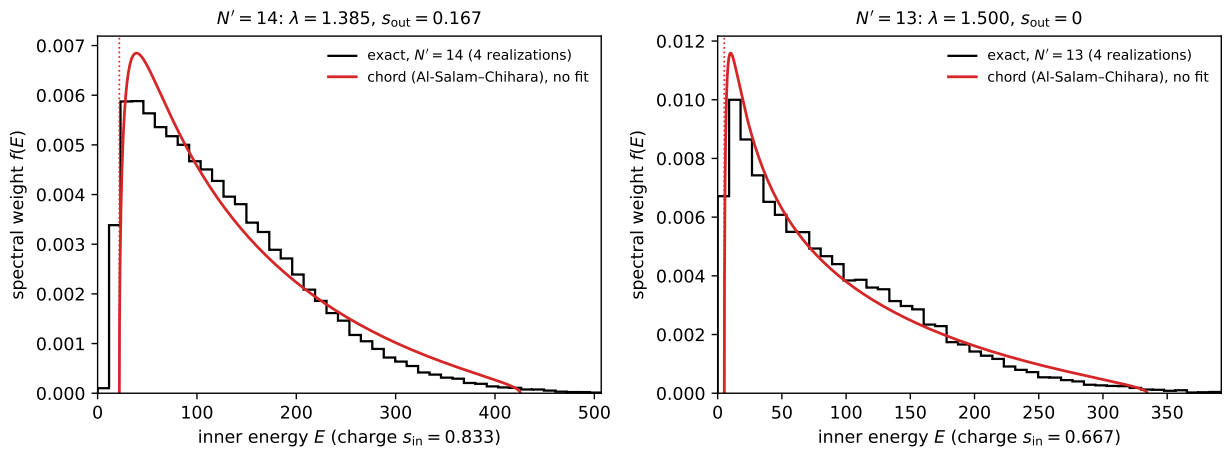


Figure 1: BPS-to-continuum spectral weight of the pair operator, $f(E)$ with $E = \kappa\Lambda_{s_{\text{in}}}$, for the h family. Black: exact old models at $N' = 14$ and 13 (four realizations; weight $\mathcal{V}_{nn}/(b\,c_0)$ of each inner level, binned). Red: Theorem 1 at $\lambda = 2p^2/N$, κ from the first moment, no fit. The dotted line is the double-scaled gap. The bulk agrees. The finite-p spectrum is broader and leaks below the gap, which matters for the $1/E$-weighted dressing cost.

Proof. We have $\Lambda_{s_{\text{in}}} = q^{-1/2}(a+a^{-1}-2x)$, so the integral is $\frac{q^{1/2}}{2}F(x_a)$, where $F(z) = \int d\mu(x)/(z-x)$ is the Stieltjes transform of the Al-Salam–Chihara measure μ and $x_a = (a+a^{-1})/2 > 1$. Let p_n be the monic orthogonal polynomials. They satisfy $xp_n = p_{n+1} + \frac{a+b}{2}Q^n p_n + \gamma_n p_{n-1}$ with $Q = q^2$ and $\gamma_n = \frac{1}{4}(1-Q^n)(1-abQ^{n-1})$ [7]. Let q_n be the second-kind functions, which form the minimal solution of the same recursion. The Casoratian $W_n = p_n q_{n-1} - p_{n-1}q_n$ obeys $W_{n+1} = \gamma_n W_n$ with $W_1 = 1$. Telescoping $q_n/p_n - q_{n+1}/p_{n+1} = W_{n+1}/(p_n p_{n+1})$ and using $q_n/p_n \to 0$ gives

$$F(z) = \sum_{n\geq 0} \frac{\gamma_1 \cdots \gamma_n}{p_n(z)\, p_{n+1}(z)}.$$

At $z = x_a$ the ${}_3\phi_2$ representation of the polynomials has the parameter $ae^{-i\theta} = 1$, so it truncates to its first term: $p_n(x_a) = (ab;Q)_n/(2a)^n$. These grow like a^{-n}, so they are dominant, as required. Substituting, the n-th term is $2a\,a^{2n}(Q;Q)_n/(ab;Q)_{n+1}$. Summing gives $F(x_a) = \frac{2a}{1-ab}\,{}_2\phi_1(Q,Q;abQ;Q,a^2)$, and $q^{1/2}a = q^{s_{\text{in}}}$. □

The ${}_2\phi_1$ agrees with direct quadrature to 10^{-15}. As $\lambda \to 0$, $(Q;Q)_n/(Q^{1+\delta};Q)_n \to \Gamma(1+\delta)n^{-\delta}$ and $1-a^2 \simeq \lambda(2s_{\text{in}}-1)$, so

$$\int_0^\pi \frac{f}{\Lambda_{s_{\text{in}}}}\, d\phi \simeq \frac{\pi}{2\sin\pi\delta}\,\lambda^{\delta-2}\,(2s_{\text{in}}-1)^{\delta-1}. \tag{4.2}$$

The integral is dominated by the near-gap (Schwarzian) region. With $c_0/\kappa \simeq \lambda/p$ at large N, the double-scaled mean dressing cost grows like $t_1 \propto \lambda^{\delta-1} \propto N^{1/3}$ for the pair operator.

Comparison with finite N [numerical]. Table 1 and Figure 2 compare t_1 with the exact old models. The chord theory reproduces the parity alternation and the size of t_1. It is low by 4–15% in the h family and by 0–6% in the g family. The deficit is largest at odd N', where the gap is small. Figure 1 shows the reason. The exact spectrum of the charged sector is broader than the double-scaled one (squared coefficient of variation 0.36–0.41 against 0.32–0.36), and it extends below

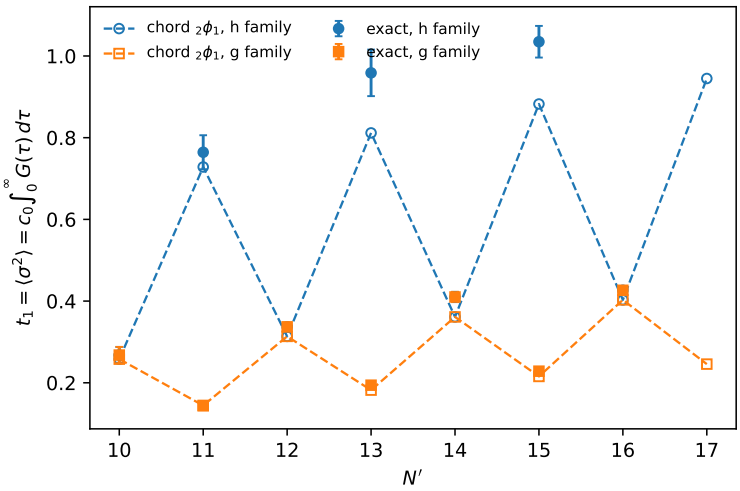


Figure 2: Mean first-order dressing cost. Filled symbols: exact (mean ± SE over 2–4 realizations). Open symbols: Proposition 1, no fit. The alternation with parity comes from the target charge: at odd N' the h family lands at $s_{\text{in}} = \frac{2}{3}$, close to the gapless sector $s_{\text{in}} = \frac{1}{2}$, and the g family at $s_{\text{in}} = 1$.

the double-scaled gap (lowest level 13.4 against 21.9 at $N' = 14$). The time integral weights this tail by $1/E$. Recalibrating λ does not help: at $N' = 14$ the exact spectrum is as broad as the broadest chord spectrum at this charge. This is a finite-p effect ($p = 3$).

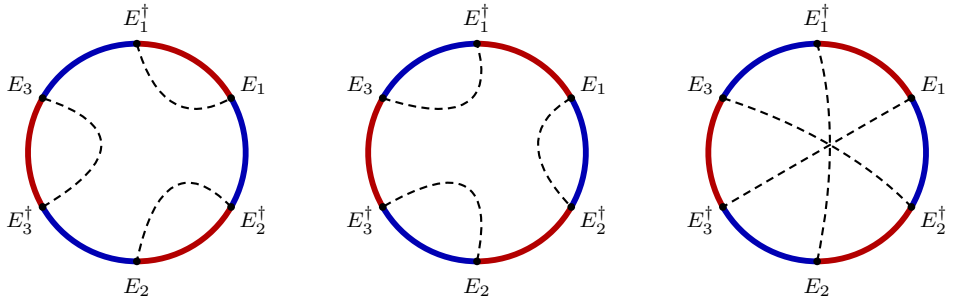
Table 1: Mean dressing cost t_1: exact (mean ± SE over realizations) and Proposition 1. Also listed: the fraction of $\|S\|_F^2$ on the diagonal of $S = G^{1/2}\mathcal{V}G^{1/2}$ in the inner energy basis (§5), and the lowest inner energy.

N', fam.	real.	t_1 exact	t_1 chord	chord/exact	diag. frac. of S	$E_{\min}$ exact / chord gap
10, h (= g)	4	0.268 ± 0.020	0.258	0.96	0.9980	15.3 / 16.0
11, h	4	0.764 ± 0.042	0.728	0.95	0.9969	3.4 / 4.4
11, g	4	0.144 ± 0.010	0.144	1.00	0.9885	40.6 / 38.4
12, h (= g)	4	0.337 ± 0.007	0.314	0.93	0.9967	14.1 / 18.9
13, h	4	0.959 ± 0.057	0.812	0.85	0.9948	2.8 / 5.2
13, g	4	0.194 ± 0.008	0.182	0.94	0.9862	35.4 / 45.1
14, h (= g)	4	0.410 ± 0.012	0.361	0.88	0.9952	13.4 / 21.9
15, h	3	1.035 ± 0.039	0.882	0.85	0.9936	3.1 / 5.9
15, g	3	0.228 ± 0.005	0.216	0.94	0.9853	36.9 / 51.8
16, h (= g)	1	0.426	0.403	0.94	0.9942	14.7 / 24.9

5 The whole spectrum: matter-chord diagrams

5.1 Wick contractions are matter-chord diagrams

The couplings η are Gaussian with $E\,\eta_I \bar{\eta}_J = 2\delta_{IJ}$. The moments $E_\eta \operatorname{tr} \mathcal{D}^m$ (with $\operatorname{tr} = \frac{1}{b}\operatorname{Tr}$) are therefore sums over the $m!$ ways $\pi \in S_m$ of contracting each E_i with some $E^\dagger_{\pi(i)}$. Each contraction is a disk whose boundary carries $2m$ insertions and $2m$ segments (Figure 3). The segments alternate between zero-temperature segments P at charge s_{out} and integrated segments $\int d\tau\, e^{-\tau H_0}$ at charge s_{in}, and the m contractions are drawn as matter chords. A contraction is *uncrossed* if no two matter



(a) uncrossed, $\pi = \mathrm{id}$ (b) uncrossed, $\pi = (123)$ (c) crossed, $\pi = (132)$

Figure 3: Three of the six Wick contractions of $E_\eta \operatorname{tr} \mathcal{D}^3$, drawn as matter-chord diagrams (dashed). Blue segments are zero-temperature (BPS) segments; red segments are the time-integrated propagation in the charged sector. (a) Three G-faces of size 1 give t_1^3. (b) One G-face of size 3 gives t_3. (c) The single crossed contraction.

chords cross, which happens exactly when π is a non-crossing permutation; there are Catalan(m) of them. The faces of an uncrossed diagram are of two kinds. P-faces contain only zero-temperature segments, and G-faces contain only integrated segments. The G-faces correspond to the cycles of π.

5.2 Uncrossed diagrams sum to a free compound Poisson law

Proposition 2 (Free compound Poisson)**.** *[derived under the chord rules] Assume the chord rules for uncrossed matter correlators: each face of an uncrossed diagram carries a single energy, and each matter chord between faces at energies ϕ, ϕ' contributes the two-point kernel of §3 (BINT [6]; BLY [4, §5] for $\mathcal{N} = 2$). Then the uncrossed part of $E_\eta \operatorname{tr} \mathcal{D}^m$ equals $\sum_{\pi \in NC(m)} \prod_{G\text{-faces } F} t_{|F|}$, with*

$$t_k = \frac{1}{b} \operatorname{Tr} S^k, \qquad S = G^{1/2} \mathcal{V} G^{1/2} \text{ diagonal in energy}, \qquad S(\phi) = g(\phi) = \frac{b\, c_0}{D_{\mathrm{in}}} \frac{A(\phi)}{\kappa \Lambda_{s_{\mathrm{in}}}(\phi)}, \tag{5.1}$$

where $\mathcal{V} = E_\eta[E\, P\, E^\dagger]$ and $D_{\mathrm{in}} = \binom{N}{F+2}$. This is the moment–cumulant formula of free probability with free cumulants $\kappa_k(\mathcal{D}) = t_k$. Without crossed diagrams, $\mathcal{D}$ is therefore distributed as a free compound Poisson law: rate $r = e_{F+2}/b_F$, and jump distribution $g(\phi)$ with ϕ drawn from the normalized density ρ_B. The same holds for the g family.

Sketch.

- A P-face is a region in which every boundary segment is an infinitely long zero-temperature evolution at charge s_{out}. That projects the face's single energy onto the unique zero mode $\Psi_{s_{\mathrm{out}}}$. The face is therefore rank one: it contributes the BPS density b and no further dependence on the BPS state.
- A G-face with k integrated segments is bounded by k matter chords, each joining it to a P-face. Each chord contributes $A(\phi)$ by Theorem 1, and each segment contributes $\int d\tau\, e^{-\tau\kappa\Lambda} = 1/(\kappa\Lambda)$.
- Summing the face energy over the $\mathcal{B}$-family states, $D_{\mathrm{in}} \rho_B(\phi) d\phi$, gives $\operatorname{Tr} S^k$ with S as in (5.1). The normalization uses $E_\eta \langle h | E^\dagger E | h \rangle = c_0$ and $\int f = 1$.
- Faces of an uncrossed diagram are G-faces for the cycles of π and P-faces for the Kreweras complement. With the normalized trace this is the free moment–cumulant formula.

□

The law in Proposition 2 is also that of $Z^\dagger S Z$, where Z is an $e \times b$ matrix of i.i.d. Gaussians with variance $1/b$. Its atoms, of mass $1 - r$ when $r < 1$, are the refined rank bound of [2, §8]. Two

exact finite-N identities hold without any chord input [derived]:

$$E_\eta \operatorname{tr} \mathcal{D} = \tfrac{1}{b} \operatorname{Tr} S = \operatorname{tr} \bar{\mathcal{D}}, \qquad E_\eta \operatorname{tr} \mathcal{D}^2 = \operatorname{tr} \bar{\mathcal{D}}^2 + \tfrac{1}{b} \operatorname{Tr} S^2, \qquad \bar{\mathcal{D}} \equiv E_\eta \mathcal{D} = 2 \textstyle\sum_I P \psi_I G \psi_I^\dagger P. \quad (5.2)$$

Both are reproduced by the separable model $X = S^{1/2} Z C^{1/2}$ with $C = \bar{\mathcal{D}}/t_1$. The chord theory says $C = \mathbb{1}$ (P-faces are rank one). It also says S is diagonal in energy, since G-faces carry one energy. At finite N the second statement holds to 98.5–99.8% of $\|S\|_F^2$ (Table 1). The first fails: the eigenvalues of $\bar{\mathcal{D}}$ spread by 12–42% (§8).

5.3 Crossed diagrams

A crossed diagram needs the matter–matter crossing (R-matrix) of the $\mathcal{N} = 2$ chord theory. BINT [6] constructed this for $\mathcal{N} = 0$. BLY set up the $\mathcal{N} = 2$ one-particle Hilbert space but did not evaluate a crossed correlator. We therefore measure the crossed contribution directly [numerical]. Take three independent Gaussian coupling vectors α, β, γ and $X_v = G^{1/2} E_v V_b$. Then

$$\tfrac{1}{b} \operatorname{Tr} \left[(X_\gamma^\dagger X_\alpha)(X_\beta^\dagger X_\gamma)(X_\alpha^\dagger X_\beta) \right]$$

is an unbiased estimator of the crossed term of $E_\eta \operatorname{tr} \mathcal{D}^3$ alone, because each vector is contracted only with itself. The same construction for the identity and full-cycle contractions reproduces their exact values $\operatorname{tr} \bar{\mathcal{D}}^3$ and $\frac{1}{b} \operatorname{Tr} S^3$, which validates the estimator. Table 2 shows that the crossed term is between -0.5% and $+0.2\%$ of the third moment and shrinks with N'.

Table 2: The crossed matter-chord diagram as a fraction of $E_\eta \operatorname{tr} \mathcal{D}^3$ (two realizations per size; statistical errors $\pm 0.02\%$).

N'	10	11	12	13	14
h family	-0.44%	$+0.04\%$, -0.02%	-0.14%, -0.17%	$+0.07\%$, $+0.04\%$	$+0.12\%$, $+0.11\%$
g family	-0.43%, -0.51%	-0.37%, -0.39%	-0.15%, -0.16%	-0.10%, -0.08%	$+0.09\%$, $+0.13\%$

6 Gravity versus exact at $N' = 10$–16

6.1 An error budget

For each realization we compute five versions of the first-order spectrum. Each step changes one ingredient.

- *exact*: the spectrum of $\mathcal{D}$ for the realization's own η (identical to the first-order data of [2]), and its average over 24 further independent draws of η;
- *planar, exact inputs*: the separable model with the exact S and $C = \bar{\mathcal{D}}/t_1$. This keeps the exact first two moments and drops everything else;
- *factorized*: the same with $C = \mathbb{1}$ (rank-one P-faces);
- *chord*: the population of S from (5.1), with $C = \mathbb{1}$;
- *chord, rescaled*: the chord population rescaled to the exact t_1.

Table 3 and Figure 4 give the result, with the decoder's W for reference.

The budget breaks down as follows [numerical].

Table 3: First-order heterogeneity $W^{(1)}$ (b-weighted over both families) and $T_2^{(1)}$. "Exact" is the mean over the 13 realizations of [2] (6 at $N' = 15$, 2 at $N' = 16$). The models use 4, 4, 4, 4, 4, 3 and 1 realizations respectively. Every model column is a mean over realizations with standard error ≤ 0.0005. The last column is the decoder's W [2, Tables 5–6]; *exact decoder on the two pre-registered $N' = 16$ realizations, $W^{(1)}$ plus the measured residual.

N'	$W^{(1)}$ exact	planar, exact inputs	factorized	chord	chord/exact	$T_2^{(1)}$ exact / chord	W^{dec}
10	0.0587	0.0628	0.0623	0.0612	1.043	0.702 / 0.691	0.0666
11	0.0897	0.0947	0.0947	0.0967	1.078	0.625 / 0.608	0.0891
12	0.0838	0.0890	0.0893	0.0901	1.075	0.637 / 0.615	0.0897
13	0.1107	0.1161	0.1170	0.1202	1.086	0.565 / 0.545	0.1083
14	0.1045	0.1098	0.1105	0.1123	1.075	0.568 / 0.552	0.1085
15	0.1264	0.1314	0.1325	0.1365	1.080	0.525 / 0.494	0.123
16	0.1203	0.1242	0.1251	0.1281	1.065	0.535 / 0.502	0.124*

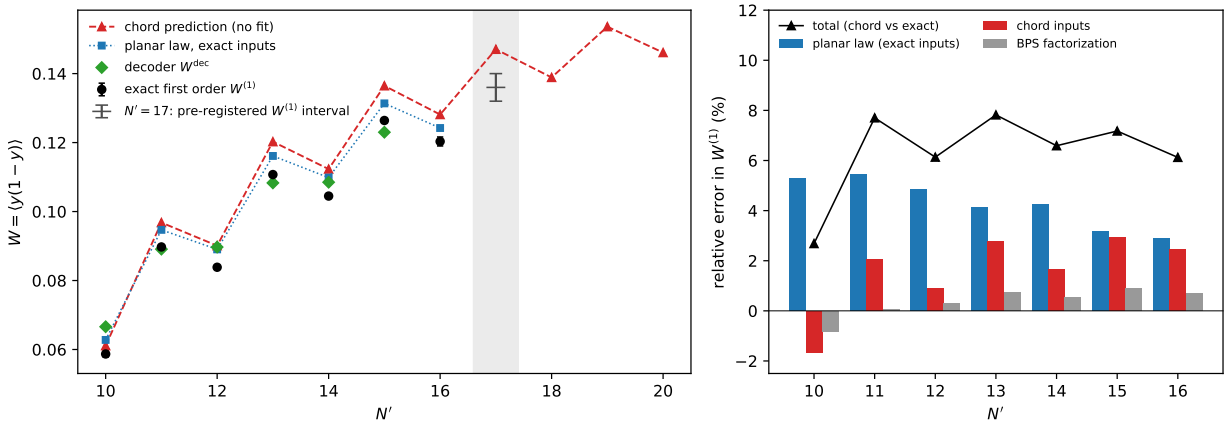


Figure 4: (a) First-order heterogeneity: exact (black), chord prediction with no fit (red), planar law with exact inputs (blue), and the decoder's W (green). The shaded band is the pre-registered $N' = 17$ test (§7). (b) Error budget of the chord prediction relative to the exact first order: planar law, chord inputs and BPS factorization, with their total.

1. **The planar law with exact inputs is 3–5% high in** W: +5.4% at $N' = 10$, falling to +3.2% at 15 and +2.9% (one realization) at 16. It is accurate for T_2. The difference sits at small σ^2, which W weights heavily ($w(s)$ peaks at $s = 0.17$; Figure 5). The exact spectrum has a softer lower edge: more weight at $\sigma^2 \lesssim 0.03$, and at $N' = 13$–14 also the deterministic atom excess of [2, §8]. This is not a crossed-diagram effect (Table 2). Nor is it the inner-level structure: at $N' = 10$, giving every inner energy eigenstate its own independent couplings (keeping each row's covariance exactly) restores the planar edge (1% quantile 0.025–0.037 against 0.015–0.027 exact). The softness comes from the fact that all matrix elements share the same $\binom{N}{2}$ couplings. Its decrease with N' is consistent with that.

2. **The chord inputs add 1–3% to** W and lower $T_2^{(1)}$ by 2–6%. This is the finite-p spectrum of §4. Rescaling the chord population to the exact t_1 changes W by at most 1% (for example 0.1376 instead of 0.1365 at $N' = 15$). W is insensitive to the overall scale and sensitive to the shape.

3. **BPS factorization costs less than 1% in** W, although $\bar{\mathcal{D}}$ is far from $\propto \mathbb{1}$. At these sizes

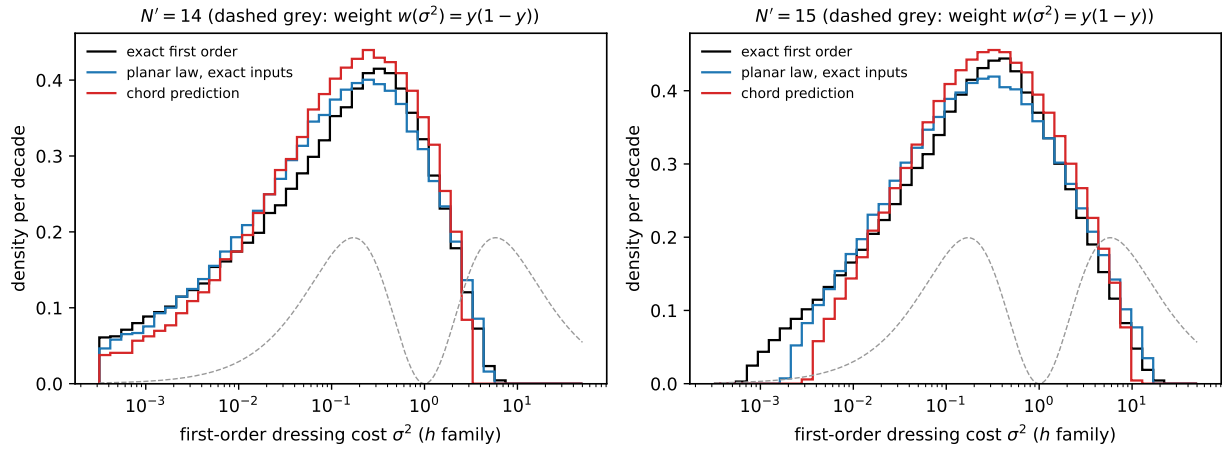


Figure 5: Distribution of first-order dressing costs (h family, one realization, several η draws). The exact spectrum (black) has more weight at the smallest σ^2 than the planar law with exact inputs (blue) and the chord prediction (red). The dashed grey curve is the weight $w(\sigma^2) = y(1-y)$ that enters W.

$r = e/b \simeq 0.4$–1.3, so the intrinsic spread of the free law dominates.

At $N' = 10$ the first two errors partly cancel. Elsewhere they add, to +7–9%.

6.2 Consequences for the decoder

Replace $W^{(1)}$ by its chord value in the prediction $T_4^{\rm dec} \approx T_2^{\rm dec} - W^{(1)}$ of [2]. The errors in T_4 become $+0.005$, -0.008, -0.000, -0.012, -0.004, -0.014 for $N' = 10$–15. With BLY's chord T_2 as well [2, §9.3], the prediction contains no input from the N'-mode model and no fit. Its errors are $+0.016$, -0.004, $+0.004$, -0.015, -0.006, -0.022, at most 4% of T_4. The localization predicted from the chord W is 0.038, 0.030, 0.041, 0.031, 0.043, 0.032, against the measured 0.032, 0.038, 0.042, 0.043, 0.047, 0.045. That is within 1–8% at $N' = 12$ and 14, and 20–30% low at odd N'. At odd N' the chord excess in $W^{(1)}$ adds to the negative first-order residual $W^{\rm dec} - W^{(1)}$ of [2].

7 A pre-registered test at $N' = 17$

[pre-registered] The next size is $N' = 17$: an old model with $N = 16$ modes, $F = 8$, $b_8 = 4374$, $b_7 = 2187$ and $e_{10} = 7449$ inner exact forms. Before computing anything at this size, we wrote down predictions in `results/data/chord_dressing_2026-10-14/prereg_N17.json` and committed them (commit `097cd6e`). They use only the chord theory and the calibration at odd $N' = 11$, 13, 15. The parameter-free chord values are $W^{(1)} = 0.1470$, $T_2^{(1)} = 0.4520$, and $t_1 = 0.945$ (h) and 0.246 (g). At odd N' the ratio exact/chord of $W^{(1)}$ has been 0.927, 0.921, 0.926. It is stable because two errors move in opposite directions: the planar error decreases (+5.5, +4.1, +3.2%) while the chord-input error increases (+2.1, +2.8, +3.0%). Table 4 lists the predictions. The protocol fixes the realizations (seeds 9000 and 9001, couplings as in [2]), the command, and the summary statistic: the mean over the two realizations of the first-order spectra for the realizations' own couplings.

Status. The test has not been run at the time of writing. It is the first computation to do next. The memory-lean code peaked at about 2 GB at $N' = 16$. The dominant arrays grow by factors of 4–7 from $N' = 16$ to 17, so we expect about 10 GB, to be measured under the watchdog.

Table 4: Pre-registered predictions for $N' = 17$ (`prereg_N17.json`). "Chord" is the parameter-free value; the intervals combine it with the odd-N' calibration, the realization-to-realization scatter and the observed trends.

	Quantity	Chord	Pass interval
P1	$W^{(1)}$, exact first order	0.1470	[0.132, 0.140]
P2	$W^{(1)}_{\text{chord}}/W^{(1)}_{\text{exact}}$	–	[1.03, 1.12]
P3	$T_2^{(1)}$, exact first order	0.4520	[0.463, 0.513]
P4	t_1, h family	0.945	[0.90, 1.31]
P5	t_1, g family	0.246	[0.230, 0.290]
P6	exact first-order atoms	0	0
P7	planar-law error in W	–	[1.0, 4.5]%
P8	chord-input error in W	–	[1.5, 4.5]%
P9	diagonal fraction of S	1	≥ 0.990 (h), ≥ 0.980 (g)
P10	spread of spec $\bar{\mathcal{D}}$ (§8)	0	[0.33, 0.50] (h), [0.12, 0.20] (g)

8 BPS heterogeneity of the dressing cost

8.1 What the chord theory misses

In the chord theory every P-face is rank one. The averaged dressing operator $\bar{\mathcal{D}} = E_\eta \mathcal{D}$ is therefore proportional to the identity: every BPS state has the same mean dressing cost. At finite N this fails (Table 5, Figure 6a). The eigenvalues of $\bar{\mathcal{D}}$ spread by 22–42% in the h family and 12–16% in the g family, with no decrease from $N' = 10$ to 16. At these sizes the spread costs less than 1% in $W^{(1)}$ (§6), because the intrinsic spread of the free law dominates. That intrinsic spread shrinks once $r = e/b \gg 1$, however, and the BPS heterogeneity would then control W. Its fate decides whether the chord description survives at large N.

The heterogeneity is a late-time effect. By (2.1), $K(\tau) = P\, E_\eta[E^\dagger e^{-\tau H_0} E]\, P$ is exactly proportional to P at $\tau = 0$. Its relative spread grows from about 1.5% at $\tau E_{\text{med}} = 0.1$ to 12% at 1 and 50–75% at 10 (Figure 6b). Correspondingly, the lowest 5% of the inner levels carry 31–89% of the variance of $\bar{\mathcal{D}}$.

It is not a pair-dependent energy cost. If creating the pair (ij) simply cost a pair-dependent energy, $\bar{\mathcal{D}}$ would be a weighted sum of hole-pair projectors, $\sum_{i<j} g_{ij}\, P(1-n_i)(1-n_j)P$. A least-squares fit of this form explains only 4–18% of the variance.

8.2 A sum rule

Proposition 3 (Single-fermion sum rule)**.** *[derived + verified] Let $c_i = \iota_i$ annihilate mode i, and let $E_i = \{Q_0, c_i\} = (\iota_i \omega_0)\wedge$ be the pair-creation operator built from the old couplings $C_{i\ldots}$. On the BPS space of degree F,*

$$B_c + \sum_i P\, E_i^\dagger G\, E_i\, P = F\, P, \qquad B_c \equiv \sum_i P\, c_i^\dagger P_{F-1} c_i\, P. \tag{8.1}$$

Proof. Because ω_0 has odd degree, $\iota_i(\omega_0 \wedge \beta) = (\iota_i\omega_0)\wedge\beta - \omega_0 \wedge \iota_i\beta$, so $\{c_i, Q_0\} = E_i$. Also $\{c_i, Q_0^\dagger\} = 0$, since $Q_0^\dagger$ is a product of three annihilators. Take a harmonic α, so $Q_0\alpha = Q_0^\dagger\alpha = 0$.

- $Q_0^\dagger c_i \alpha = -c_i Q_0^\dagger \alpha = 0$, so $c_i\alpha$ is harmonic plus coexact: $c_i\alpha = P_{F-1}c_i\alpha + Q_0^\dagger\beta_i$.
- $Q_0 c_i\alpha = E_i\alpha$, so $Q_0 Q_0^\dagger \beta_i = E_i\alpha$. Since $E_i\alpha$ is exact, $\|Q_0^\dagger\beta_i\|^2 = \langle\alpha|E_i^\dagger G E_i|\alpha\rangle$.

- By Pythagoras, $\|c_i\alpha\|^2 = \|P_{F-1}c_i\alpha\|^2 + \langle\alpha|E_i^\dagger G E_i|\alpha\rangle$.

Summing over i with $\sum_i c_i^\dagger c_i = F$ on Λ^F gives $\langle\alpha|(\text{LHS} - F)|\alpha\rangle = 0$ for every BPS α. The operator identity follows by polarization. □

For each old site, the dressing cost of that site's own pair operator and the BPS weight of its single-fermion descendant add up to one. The added mode's E is an operator of the same kind, with independent couplings. Numerically the identity holds to 2.5×10^{-14}. The η-averaged $\bar{\mathcal{D}}$ correlates with the old-site version $\sum_i P E_i^\dagger G E_i P$. Its traceless correlation with B_c is therefore -0.73 to -0.93 (Table 5). The heterogeneity that the chord theory misses is thus the microstate dependence of BPS-projected single-fermion two-point functions. Microstates whose single-fermion descendants stay BPS (large B_c) are cheap to dress, and microstates whose descendants leave the BPS sector are expensive. This is the same kind of quantity as the decoder's own heterogeneity.

8.3 Incoherent and coherent parts

Write $\bar{\mathcal{D}} = \sum_n M_n/E_n$, where $M_n = 2\sum_I P\psi_I|n\rangle\langle n|\psi_I^\dagger P$ is the contribution of inner level n. Each M_n has rank at most $\binom{N}{2}$. If the M_n were oriented at random (Haar) relative to each other, free probability would give the eigenvalue variance $\sum_n[\operatorname{tr} M_n^2 - (\operatorname{tr} M_n)^2]/E_n^2$. We call this the *incoherent* estimate; it is computed exactly from the Gram matrices of the vectors $P\psi_I|n\rangle$.

Table 5 compares it with the observed spread. We look at the lowest 5% of levels, where the variance lives.

- The incoherent estimate is exact at $N' = 10$ and falls increasingly short: observed/incoherent $= 1.00, 1.09, 1.15, 1.37, 1.41, 1.72$ for $N' = 10$–15 (h).
- The coherent share $1 - (\text{incoherent/observed})^2$ grows from 0 to 0.66 (h) and to 0.54 (g) (Figure 6c).
- The effective number of contributing levels grows from 21 to 365, and the incoherent part shrinks as it should. The total stays flat because the coherent part takes over.

For the whole operator the comparison is complicated by the exact sum rule $\sum_n M_n = c_0 P$, which makes the energy bins anti-correlated.

[interpretation] Two mechanisms are at work.

- An *incoherent* one: a finite number of low-lying levels, each coupled through a low-rank channel. It decays like $1/N_{\text{eff}}$ and is harmless for the chord theory at large N.
- A *coherent* one: the same BPS microstates are IR-heavy across many low-lying levels, and they are the ones with small single-fermion BPS weight. It does not decay in our window.

Its large-N fate at fixed p is the same open question as that of localization itself.

9 Discussion

9.1 What we have learned

The dressing operator of the uplift problem is computable in the chord theory of $\mathcal{N} = 2$ SYK, and three of its ingredients are exact. Theorem 1 puts BLY's BPS-to-BPS and LMRS's BPS-to-continuum zero-temperature two-point functions into a single Al-Salam–Chihara measure. The time integral is a ${}_2\phi_1$. The uncrossed matter-chord diagrams sum to a free compound Poisson law.

Table 5: BPS heterogeneity of $\bar{\mathcal{D}}$ (two realizations per size; at even N' the g family equals the h family by Hodge duality). Columns: relative spread of spec $\bar{\mathcal{D}}$, observed and incoherent estimate; the same for the lowest 5% of levels; coherent share of that variance; variance share of the lowest 5% of levels; effective number of contributing levels; traceless correlation of $\bar{\mathcal{D}}$ with the single-fermion BPS weight B_c; spread of spec B_c.

N', fam.	spread	incoh.	low 5%: obs / incoh.	coh. share	var. share	N_{eff}	corr$(\bar{\mathcal{D}}, B_c)$	spread B_c
10, h	0.216	0.356	1.75 / 1.75	0.00	0.31	21	−0.84	0.146
11, h	0.396	0.389	1.27 / 1.16	0.17	0.71	37	−0.73	0.336
11, g	0.120	0.271	1.04 / 1.02	0.04	0.37	28	−0.90	0.068
12, h	0.253	0.283	1.05 / 0.91	0.25	0.63	72	−0.86	0.164
13, h	0.411	0.306	1.00 / 0.73	0.48	0.83	116	−0.77	0.345
13, g	0.142	0.206	0.83 / 0.71	0.27	0.57	106	−0.91	0.077
14, h	0.246	0.218	0.89 / 0.63	0.50	0.74	252	−0.87	0.161
15, h	0.417	0.248	0.91 / 0.53	0.66	0.89	365	−0.79	0.339
15, g	0.156	0.162	0.71 / 0.48	0.54	0.76	373	−0.93	0.082

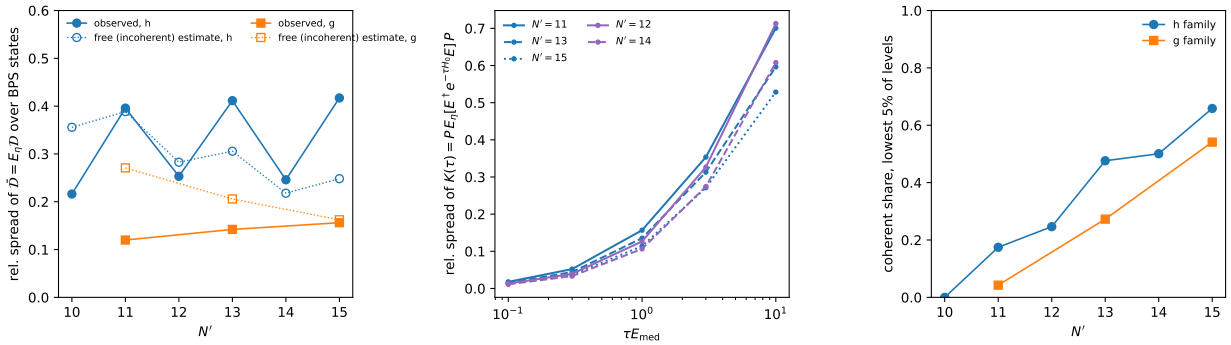


Figure 6: BPS heterogeneity of the averaged dressing cost. (a) Relative spread of the eigenvalues of $\bar{\mathcal{D}}$: observed (filled) and incoherent (free-probability) estimate (open). (b) Spread of $K(\tau)$ against Euclidean time; it vanishes at $\tau = 0$. (c) Coherent share of the variance carried by the lowest 5% of inner levels.

At the sizes we can reach, the resulting gravitational prediction of the first-order heterogeneity is accurate to 4–9% with no fitted parameter, and the error budget accounts for the rest. Crossed diagrams are negligible. The planar law has a slightly too hard lower edge, because all matrix elements share $\binom{N}{2}$ couplings. The finite-p spectrum is broader than the double-scaled one.

The physics the chord theory misses at these sizes is BPS heterogeneity. An exact sum rule ties it to single-fermion BPS-to-BPS weights. That is the localization question again, now in a different guise.

9.2 Beyond accessible sizes

Within the double-scaled model, the chord prediction extends to any size. With r the inner-to-BPS dimension ratio, the predictions are:

N'	17	18	19	20	24	30	40
$W^{(1)}_{\text{chord}}$	0.147	0.139	0.154	0.146	0.153	0.152	0.140
$r_h = e/b$	1.7	1.8	2.2	2.4	4.1	8.9	32
spread of the law, $\sqrt{t_2}/t_1$ (h)	1.64	1.35	1.59	1.29	1.20	1.10	0.99

$W^{(1)}_{\text{chord}}$ peaks near 0.15 at $N' \approx 24$–30 and then decreases slowly. Two effects compete: the free law concentrates as r grows exponentially, while t_1 grows like $N^{1/3}$. [interpretation] These are predictions of the double-scaled model, not of $p = 3$ SYK, for three reasons.

- $\lambda \to 0$ at fixed p is the Schwarzian regime of DSSYK, whose coupling ($\alpha_S = 1/8p^2$) differs from that of fixed-p SYK (LMRS).
- They are first order only.
- They omit the coherent BPS heterogeneity of §8, which would take over once $r \gg 1$.

9.3 Open problems and next steps

1. **Run the pre-registered $N' = 17$ test** (§7), and complete the stopped $N' = 16$ runs (second realization of the decomposition; heterogeneity analysis).
2. **The coherent BPS heterogeneity.** Follow its share beyond $N' = 16$. Relate it directly to the decoder's localization: both B_c and Pn_eP are BPS-projected single-fermion quantities.
3. **The soft lower edge.** An operator-valued free-probability treatment, using the exact covariance of $X = \sum_I \eta_I G^{1/2} \psi_I^\dagger P$ with its few couplings, might capture it.
4. **The $\mathcal{N} = 2$ matter–matter crossing.** It is needed for crossed diagrams here and for BLY's zero-temperature OTOC. Here it is not the bottleneck ($\leq 0.5\%$).
5. **Beyond first order.** Express the self-dressing and mixing steps of [2] in chord language.

A Verification summary

Every analytic statement was checked against an independent implementation, using the tests in `tests/test_chord_dressing.py` (8 tests; the earlier test suites also pass).

- Ground-state amplitudes (3.3): equal to the q-Hermite closed form and to the zero mode of the transfer matrix, to 10^{-14}.
- The BBNR sign remark: direct action of T on $\sum_n \alpha_n b_n$ (relative error $< 10^{-9}$).
- Theorem 1: moments $\int f \Lambda^k$, $k = -2, \ldots, 3$, against the transfer matrix, to 10^{-15} (test tolerance 10^{-8}), at four parameter points; $f = \rho_B A$ and $\int f = 1$.
- Corollary 1: point mass $= 1 - \int f =$ engine zero-mode weight $=$ BLY (4.14), to 10^{-12}.
- Proposition 1: ${}_2\phi_1$ against quadrature, to 10^{-15}.
- Finite N:
  - the lean code's first-order spectrum equals `OldModel`'s (`src/uplift.py`) to 10^{-9};
  - $\operatorname{Tr} G\mathcal{V} = \operatorname{Tr} \bar{\mathcal{D}}$;
  - the scatter construction of $\mathcal{V}$ equals the dense sum to 10^{-12};
  - the second-moment identity (5.2) holds within sampling errors;
  - the crossing estimator reproduces the exact identity and full-cycle terms.
- Proposition 3: to 2.5×10^{-14} at $N' = 10$–13.
- The committed scripts reproduce the exploratory computations exactly (same seeds).

B Numerical methods and reproducibility

Code.

- `src/dssyk_n2.py`: closed forms (Theorem 1, Corollary 1, Proposition 1).
- `src/chord_transfer.py`: transfer-matrix chord engine.
- `src/chord_dressing.py`: chord predictions, finite-N dressing machinery, crossing estimator, sum rule.

Scripts (with command-line parameters).

- `scripts/run_chord_dressing.py` (`--Np`, `--seeds`, `--neta`): error decomposition.
- `scripts/run_chord_crossing.py`: crossed diagram.
- `scripts/chord_dressing_predict.py`: chord predictions.
- `scripts/analyze_bps_heterogeneity.py`: §8.
- `scripts/plot_chord_dressing.py`: figures and tables.

Data and parameters. Everything is in `results/data/chord_dressing_2026-10-14/`, together with the driver scripts `run_calibration.sh`, `run_crossing.sh` and `run_heterogeneity.sh`, logs and the memory-watchdog log.

- Realizations: seeds $9000 + k$, with couplings exactly as in [2]. Seeds 9000–9003 for $N' = 10$–14, 9000–9002 at 15, 9000 at 16.
- η draws: 24 ($N' \leq 14$), 12 ($N' = 15$), 6 ($N' = 16$).
- Separable models: 3 draws each.
- Crossing estimator: 2000, 1500, 1000, 600, 300 samples for $N' = 10$–14.
- Chord predictions: $b_{\text{eff}} = 1500$, 3 draws (finite-b effects in W below 10^{-4}).

Memory. All runs used the watchdog (`scripts/memwatch.py`). Peaks: about 2 GB for the calibration (through $N' = 16$), 0.2 GB for the crossed diagrams, 0.8 GB for the predictions, and under 2 GB for the heterogeneity analysis.

Not completed. Two runs were stopped before finishing when this report was finalized: the second $N' = 16$ realization of the decomposition, and the $N' = 16$ heterogeneity analysis. The exploratory version of the first (seed 9001) gave $W^{(1)} = 0.1190$ for its own couplings (0.1182 averaged over 6 draws), 0.1240 for the planar law with exact inputs and 0.1281 for the chord prediction, consistent with the realization shown.

Commits. Code, data and pre-registration: `097cd6e`. This report: the following commit on branch `t4-fourarc`.

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